Paper Note of Manifold Learning for Molecular ML

* (i) AtomSurf : Surface Representation for Learning on Protein Structures (Submitted In NeurIPS 2023)

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- In this paper, we investigate representing proteins as 3D mesh surfaces and incorporate them into an established representation benchmark（Atom3D).
- Our first finding is that **despite promising preliminary results, the surface representation alone does not seem competitive with 3D grids**. Building on this, we **introduce a synergistic approach, combining surface representations with graph-based methods**, resulting in a general framework that incorporates both representations in learning.

1. Introduction

- Structural bioinformatics data is becoming available at an unprecedented pace.
- However, structural biology data displays certain additional geometric properties compared to images or sequences. For instance, **biological systems lack a canonical orientation due to the insignificance of gravity.**
- Selecting a suitable mathematical representation of protein structures (see Figure 1) and devising an effective learning method compatible with the chosen representation. **The optimal pairing of representation and learning method remains a complex task.**
- However, approaches based on the surface representation have followed the initial MaSIF paper validation [1], and hence have never been directly compared to other representations in the context of a single well-established benchmark.
- Beyond comparing the use of a single representation for proteins, a few papers mention using several representations simultaneously. **However, these methods still rely on the flexibility of the graph representation to accommodate different representations, losing the specific properties of surface methods.**
- Contribution
    - In our study, we bridge this gap by applying DiffusionNet to the Atom3d benchmark, **offering a rigorous comparison of the surface representation against other representations for the first time and introducing a batched implementation of DiffusionNet along the way。**
    - Our explorations extend to marrying surface architectures with graph methods, evaluating multiple strategies, and **culminating in the proposal of a hybrid representation technique that sets new standards in the benchmark.**

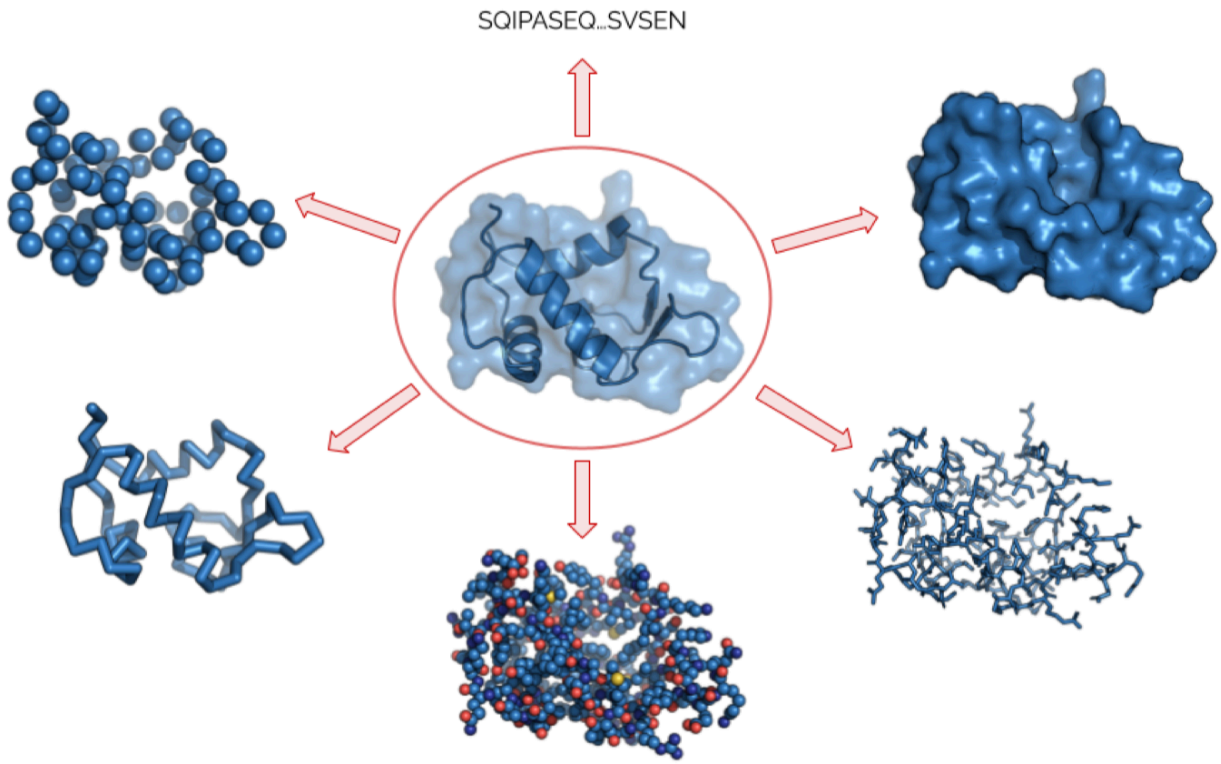

Figure 1: This figure illustrates the diverse mathematical objects used to represent a protein structure, ranging from sequences to atom-level and residue-level point clouds, graphs, and molecular surfaces. Effective machine learning for protein structures hinges on selecting the appropriate mathematical representation, followed by a compatible machine-learning technique. This dual-layered modeling approach underscores the complexity of extrapolating performance solely based on machine learning benchmarks in protein structure tasks.

[1] Gainza P, Sverrisson F, Monti F, et al. Deciphering interaction fingerprints from protein molecular surfaces using geometric deep learning[J]. Nature Methods, 2020, 17(2): 184-192.

2. Methods

2.1. Single Representation Learning

- **(Almost same as HMR)** We start by deriving the protein surface using MSMS, gradually augmenting the vertex density until a minimum of 128 vertices is obtained. We then apply quadratic decimation to reduce the size of the largest resulting meshes. **With the mesh refined, we subsequently precompute the Laplacian operator along with its 128 eigenvectors using the cotangent Laplace-Beltrami decomposition.**
    - **For methods relying on surfaces, we project the graph's initial node features onto the vertices using a Gaussian kernel.**
- Thus, the graph nodes are set to be all the atoms. **we also compute the graph representation. In the spirit of fairness, we closely adhere to the protocol set by Atom3d [2].** For the edges, we consider as neighbors all pairs of atoms below a 4.5Å radius cutoff. Finally, we use one hot vector encoding atom type as initial node features.

- **The graph encoder adopted remains consistent with that of Atom3d: it is composed of five layers of alternated Graph Convolutional Networks (GCN) [19] and BatchNorm operations.**

- We base our evaluation of the utility of surface-based learning methods on the recent approach DiffusionNet [3], which has **demonstrated state-of-the-art results on a wide range of tasks such as classification, segmentation, and shape matching,** and is particularly robust to changes in the underlying mesh structure.
- To facilitate learning, **we introduce a batched implementation of DiffusionNet.**

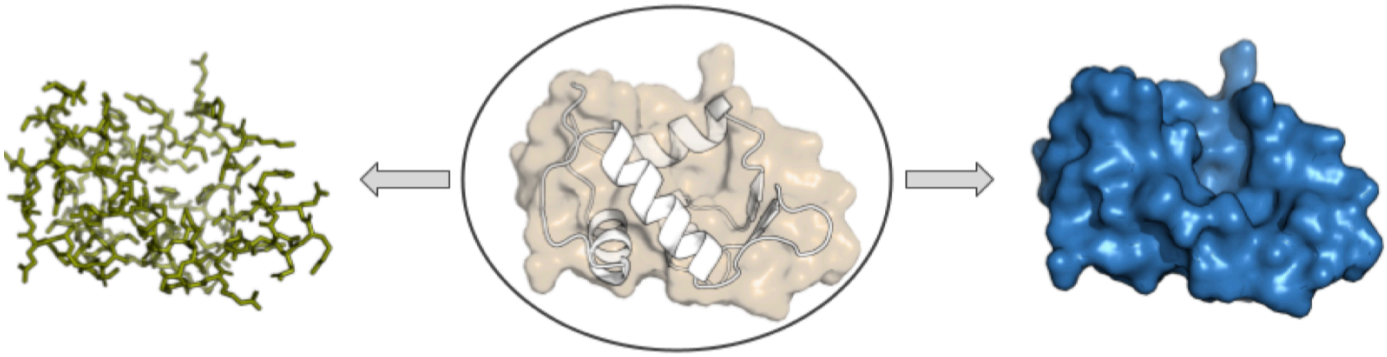

Figure 2: The MDM2 protein (pdb `1ycr`) is represented as a graph (green) and as a surface (blue). For readability, the graph displayed here relies on chemical connectivity instead of neighborhood.

[2] Townshend R J, Vögele M, Suriana P, et al. Atom3d: tasks on molecules in three dimensions. arXiv. 2020 doi: 10.48550[J]. arxiv, 2012.

[3] N. Sharp, S. Attaiki, K. Crane, and M. Ovsjanikov. Diffusionnet: Discretization agnostic learning on surfaces. ACM Transactions on Graphics (TOG), 41(3):1–16, 2022.

2.2. Multi-Representation Learning

2.2.1 Bipartite Method

- We also study the utility of combining multiple representations in a single coherent architecture.
- **The approach combines surface and graph-based representations.**
  - For this, we first **formulate a bipartite graph $G = (\mathcal{V}, E)$, with $\mathcal{V} = \mathcal{V}_g + \mathcal{V}_s$ representing the nodes in the graph juxtaposed with the vertices on the surface.**
  - We **calculate the geometric distance between node pairs spanning the two sets, establishing an edge within the bipartite graph if the distance falls below a predetermined threshold**. (A restrictive threshold might lead to a disjointed graph, whereas an overly lenient one could produce a biclique.)
  - Optionally, we also **incorporate edge features based on distances.**
  - We **designate surface and graph encoding blocks as f_θ and g_θ respectively.** Hence, the outputs after i blocks are given as $H^i = \{h_n^i | n \in V_s\}$, where $h_n^i = f_\theta^i(x_n^i)$ for a node n in V_s (or $h_n^i = g_\theta^i(x_n^i)$ for a node n in V_g).

- **Our general methodology incorporates message-passing neural networks, denoted MP^i_θ,** over the bipartite graph. By employing distinct sets, θ^i_{sg} and θ^i_{gs}, the architecture handles messages traversing from the surface to the graph and vice versa. **The primary node features of the graphs are represented as $\chi = \chi_g \cup \chi_s$, with the embeddings post i blocks labeled as χ^i, such that $\chi^i = MP^i_\theta(H^i)$.** In what follows, we drop the i superscript for simplicity and assume we are at a certain layer of the graph.
- Different forms for the message passing above result in different mixed architectures. **The model we propose is denoted bipartite**, and it is based on the following equation :

$$x_n = \alpha h_n + MP_{gs}(H)(n) + MP_{sg}(H)(n), \tag{1}$$

with $\alpha \in \mathbb{R}$. **In our proposed setting, we set α to a value of one, and use a Graph Attention Layer for the message-passing network. We discuss several ablations in the Results section.**

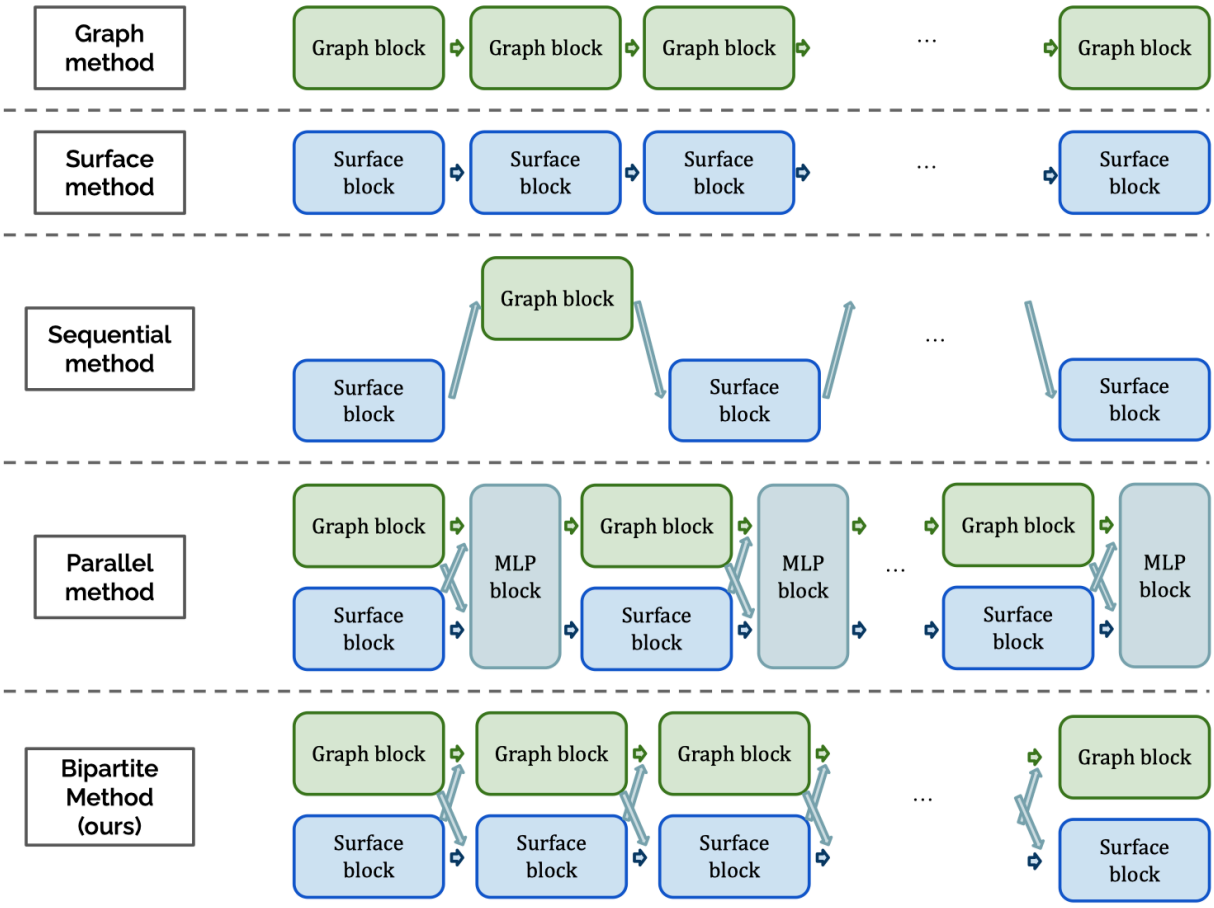

Figure 3: Different ways to leverage surface and graph information.

2.2.2 Two Baseline Strategies

The sequential setting

We alternate surface and graph encoding. We can write this in two steps. In the first step, a surface encoding is performed, then the features are projected toward the graph using message passing: $h_n = MP_{sg}(f_\theta(\chi))(n)$, resulting in intermediate graph node embeddings that will be denoted as $H^g = \{h^g_n | n \in V_g\}$. Then, we propagate these node embeddings in the graph and obtain again surface embeddings using: $\chi = MP_{gs}(g_\theta(H^g))$.

The parallel approach

The output from message passing is concatenated with the original embeddings. A multi-layer perceptron (MLP) then processes the resulting vector, which can be formulated as $x_n = MLP(h_n \| MP(H)(n))$.

3. Results

3.1 Experiment Setup

- **Protein Interaction Prediction (PIP)**

  This task aims to predict which part of a protein interacts with which part of another. Framed as a classification task, pairs of residues from two proteins are labeled 'positive' if they interact and 'negative' if they don't.

- **Mutation Stability Prediction (MSP)**

  The objective here is to determine if a mutation enhances the stability of protein-protein interaction. Given a protein-protein interaction structure and its mutated version, this classification task labels the pair as 'positive' if it exhibits increased stability.

- Protein Structure Ranking (PSR)

  PSR is a regression task and aims to assign a quality score to predicted protein structures from the Critical Assessment of Methods of Protein Structure Prediction (CASP) [20] competition.

3.2 Performance of Surface Representation

- **For the sake of fairness, we only use atom types as input and keep the same number of parameters as other methods of the Atom3d benchmark. We expect the surface representation to mostly perform well on tasks relative to interactions, and to underperform on the PSR task that also pertains to subtle changes inside the protein volume with no impact on the protein surface.**

		3DCNN	Graph	ENN	Surface
PIP	AuROC	**0.844**	0.669	-	0.837
MSP	AuROC	0.574	**0.609**	0.574	0.5
PSR	local R	**0.431**	0.411	-	0.32
	global R	**0.789**	0.75	-	0.64

Table 1: Comparison of different representations, including surface performance. The dashes in the equivariant methods' column refer to the impossibility to use the method because of memory constraints.

- We observe that the surface method consistently falls short in its performance, even on the protein-protein interaction task. Perhaps surface networks shine when supplemented with richer information. **However, when all input features are equal, surface networks are not top performers.**

3.3 Synergy in Combined Representations

- In this section, we assess the performance of our proposed method, which has the particularity of combining different representations, i.e. surface and graph.

		SOTA	Graph	Surface	Ours
PIP	AuROC	0.844	0.669	0.837	**0.876**
MSP	AuROC	0.609	0.609	0.5	**0.707**
PSR	local R	0.432	0.411	0.32	**0.452**
	global R	0.796	0.75	0.64	**0.83**

Table 2: Comparison of our proposed approach with methods relying only on graphs or surfaces and with state-of-the-art (SOTA). Our method improves current best performance on all tasks.

- Our method outperforms the state-of-the-art in all three tasks. Interestingly, its results surpass both graph-only and surface-only strategies, hinting at a synergy advantage between the two representations.

3.4 Further Analysis

3.4.1 Qualitative Results

To visualize our PIP model's predictions, we aim to display patches of probability across the protein surface. The challenge is that the PIP task operates over pairs of residues in the graph and doesn't inherently provide a global interactability score for individual residues. Given a pair of protein chains A and B, let us assume, without loss of generality, that we want to plot the interaction site of protein B. We retrieve the nodes in protein A that belong to at least one positive pair and call the sets of such nodes N_A^P. We define the interactability score of a node $I(n)$ as the indicator function of this set over graph nodes. We compute our model predicted probability $\hat{p}$ on all pairs and get a set of predictions. The final predicted interactability $\hat{I}$ of our model for a node $n_b \in N_B$ is given as $\hat{I}(n_b) = \max_{n_a \in N_A^P} \hat{p}(n_a, n_b)$.

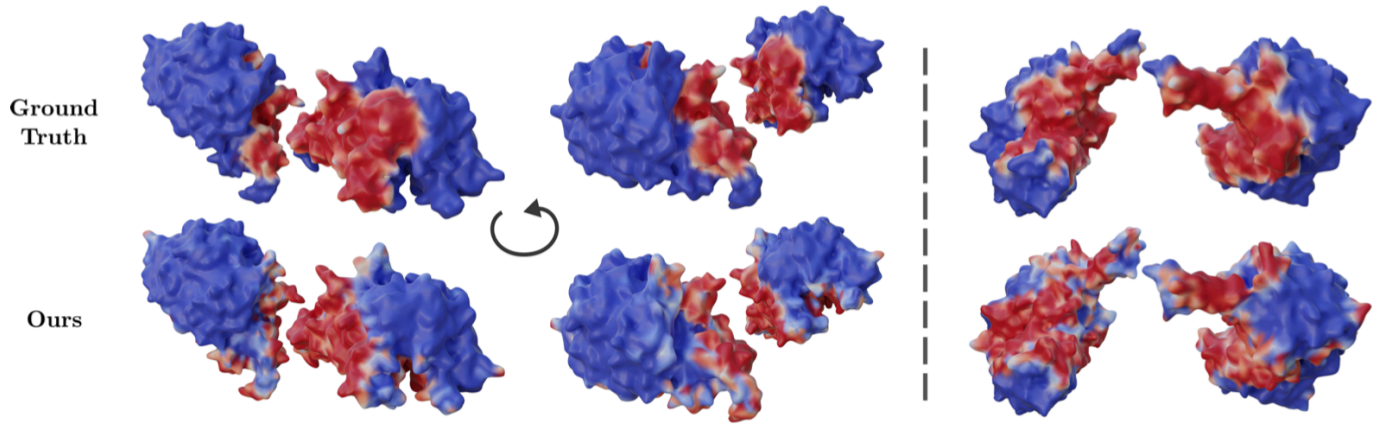

Figure 4: A qualitative view of our results. The top row is the ground truth with interaction sites in red. The bottom row displays our prediction. The two leftmost columns show the chains C and D of the protein with PDB id `2jbr` under two rotated views of 180°. On the right, the interaction between chains A and B of the system with PDB id `2ono` is shown.

3.4.2 Ablation Study

1. Holo

- **As mentioned above, the proposed mixed scenario HoloProt [28] falls into the sequential framework with just one encoding block. Hence, we add this setting in our benchmark - with an enhanced protein encoder since we replace MeshCNN [13] with DiffusionNet - and refer to it as holo.**

2. Another **major design choice is the choice of the Message Passing (MP) component**. Hence, we explore the use of three possible message-passing networks.

- Motivated by the success of DGCNN [34], in **our Att. setting, we discard the geometric notion of a neighborhood, allowing for potentially long-distance message passing.** In this setting, all nodes from the graphs attend to all vertices from the surface. To deal with the incurred computational burden, we use the recent memory-efficient Flash Attention.
- We also explore the use of more **conventional Graph Convolutional Networks (GCN setting)** and the use of **Graph ATtention networks** for our final GAT setting.
- Finally, we also **try using three or four blocks** in our networks, always adjusting the network width to keep the number of parameters constant.

Method	MP	Depth	MSP AuROC	PIP AuROC	PSR local R	PSR global R
parallel	-	-	0.55	0.5	0.39	0.77
sequential	-	-	0.609	0.855	0.319	0.71
holo	-	-	0.537	0.824	0.383	0.715
	Att.	3	0.689	0.791	0.4	0.792
	-	4	0.648	0.793	0.388	0.799
bipartite	GCN	3	0.626	0.858	0.42	0.8
	-	4	0.697	0.868	0.421	0.797
	GAT	3	**0.707**	0.859	0.434	**0.833**
	-	4	0.646	**0.876**	**0.452**	**0.83**

Table 3: Ablation study of our method. We compare different architecture designs, message passing methods and depth on our task. A detailed explanation of the different setting is available in text.

As hypothesized above, the mixed approach could simply use the surface as a way to diffuse information efficiently and at a long distance. This could explain why in this scenario, an attentive mechanism results in a performance boost.

4. Future Directions

Future directions include applying these mixed approaches to other tasks in structural bioinformatics and **optimizing the pipeline to make surface-based methods faster and less memory-demanding**. Another promising avenue is to try out more representations for learning on protein structures, especially point clouds, for which recent geometric transformers were designed with outstanding results.